

Intelligent Drug Sales Forecasting and Healthcare Insights with Artificial Intelligence

Category-wise pharmaceutical demand planning in Sri Lanka

Bope Ranasinghage Gihan Lakmal

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B.Sc. Honours in Computer Science

Faculty of Computing and Technology • University of Kelaniya

SUPERVISOR

Prof. N. G. J. Dias

From the research problem to the application

- 01** Problem & objectives
- 02** Data & methodology
- 03** System implementation
- 04** Results & explanation
- 05** Demonstration & conclusion

The problem: planning stock from changing sales

Understocking

Insufficient stock
Unmet demand
Reactive purchasing

Overstocking

Capital tied up
Storage pressure
Risk of expiry

Forecasting challenge

Different category patterns
Limited real data
Need for understandable outputs

Research question: how can historical sales support usable, category-wise forecasting?

Objectives and the delivered scope

- 01 Prepare** Organise and inspect category-wise weekly sales.
- 02 Predict** Implement statistical, boosted-tree and sequence models.
- 03 Compare** Assess MAE, RMSE, MAPE and recorded timing.
- 04 Deliver** Provide forecasts, plots and explanations in a web application.

Related work motivates a broader comparison

Study	Relevant contribution	Role in this project
Ekanayake et al., 2025 [1]	SARIMA + Flask drug-sales application	Earlier categorical forecasting work
Ke et al., 2017 [2]	Efficient gradient-boosted trees	LightGBM model comparison
Hewamalage et al., 2021 [3]	Review of recurrent forecasting models	LSTM / GRU comparison

Project extension: multi-model comparison + explanations + modular application services.

Source: Ekanayake et al. (2025), local paper; Ke et al. (2017) [2]; Hewamalage et al. (2021) [3]

The dataset: eight weekly sales series

8

drug categories

517

weeks per category

2014–2023

stored date range

50:50

real / synthetic*

12 Jan 2014 → 03 Dec 2023

One date + one sales value in each category file.

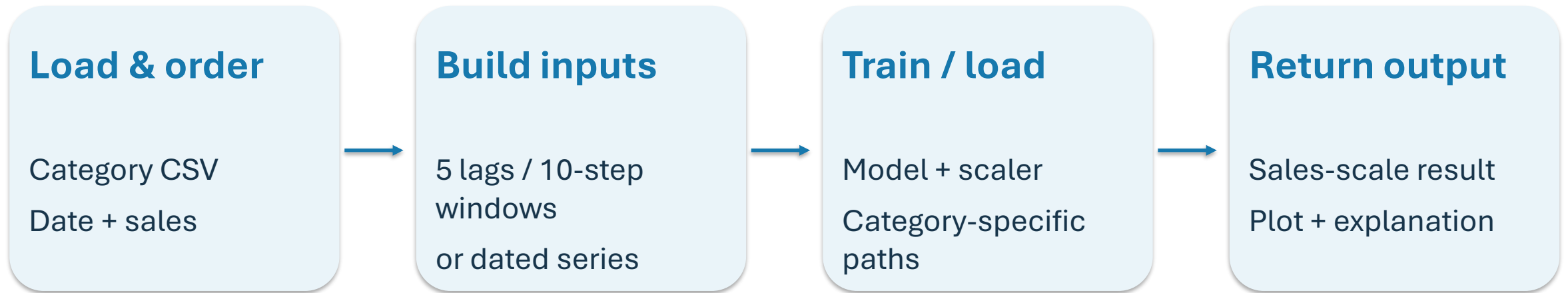
No missing cells; no duplicate dates; consistent 7-day spacing.

Source context

Two Kurunegala pharmacies
plus synthetic augmentation

***Student-confirmed approximate mix; row-level source labels and regional identifiers are unavailable.**

A model-specific preparation and forecasting pipeline



Lag input example $[y(t-1), y(t-2), y(t-3), y(t-4), y(t-5)] \rightarrow y(t)$

Implementation detail: LightGBM currently retrains locally during forecasting for artifact compatibility.

Statistical and machine learning methods

Statistical models

SARIMAX: autoregressive / seasonal structure

Evaluator: $(1,0,0) \times (1,0,0,7)$

Prophet: dated trend and seasonality inputs

Boosted-tree models

XGBoost: regression on 5 lag features

LightGBM: 5 normalised lag features

Current defaults: 31 leaves, 0.05 learning rate, 100 boosting rounds

Seasonal period 7 means seven observations in this weekly dataset; defaults are not tuned-run provenance.

Deep learning uses sequences of past sales

LSTM

10-observation windows
2 recurrent layers
64 hidden units
Dropout: 0.1

GRU

10-observation windows
2 recurrent layers
64 hidden units
Dropout: 0.1

Transformer

10-observation windows
Model dimension: 64
4 attention heads
2 encoder layers

TFT, N-BEATS and Informer are additional implementations; no comparable saved benchmark is reported.

Stacking now has measured forecasting results

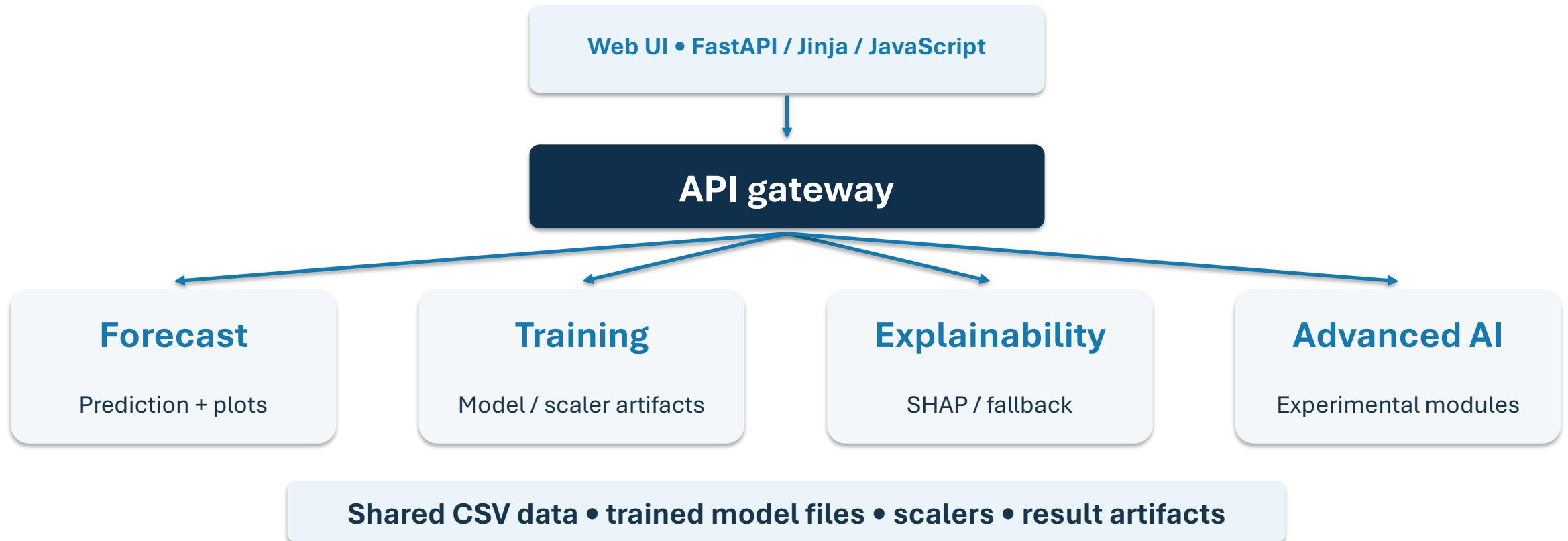
Method in the new experiment	Mean MAE	Mean RMSE
Stacking / Ridge	30.2059	36.0614
Equal average	24.2382	29.4752
XGBoost	24.7716	30.4166
LSTM	26.4038	32.9673
GRU	25.7002	30.6456
Last-value baseline	35.0069	40.7899

XGBoost + LSTM + GRU forecasts → Ridge combiner → one result per future week

New, separate chronological test: stacking beats the naive baseline, but not simple averaging.

Source: src/evaluation_results/stacking_results.json; src/models/stacking_model.py; new run, 13 Sep 2026

A modular application connects users and models



Component view: the current gateway forecast endpoint calls shared forecasting code directly.

Evaluation: distinguish the original and new experiments

MAE

Average absolute error
Lower is better

RMSE

Emphasises large errors
Lower is better

MAPE

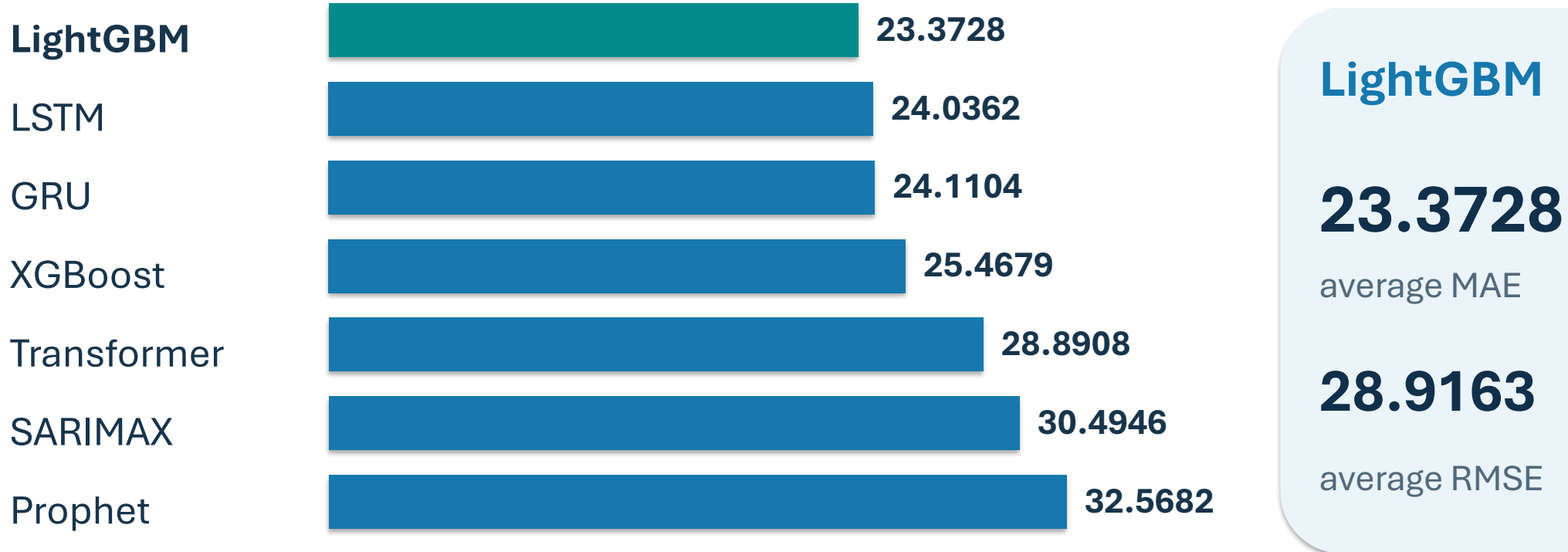
Percentage error
Sensitive to small actuals

Original benchmark + new stacking holdout

Overall values = arithmetic mean of the eight category metrics.

New stacking test: 507 training weeks + 10 untouched test weeks. Original benchmark limitations remain.

Original benchmark: LightGBM leads on absolute error



Mean category MAE • original sales scale • lower is better

Original experiment only: these numbers are not directly comparable to the new stacking test.

Original benchmark: the best model varies by category

Category	Lowest recorded MAE model	MAE
C1	GRU	9.8758
C2	Prophet	13.1088
C3	Prophet	10.2570
C4	LightGBM	72.6065
C5	LSTM	24.8986
C6	LSTM	3.1634
C7	XGBoost	26.0462
C8	LSTM	13.8154

No universal winner

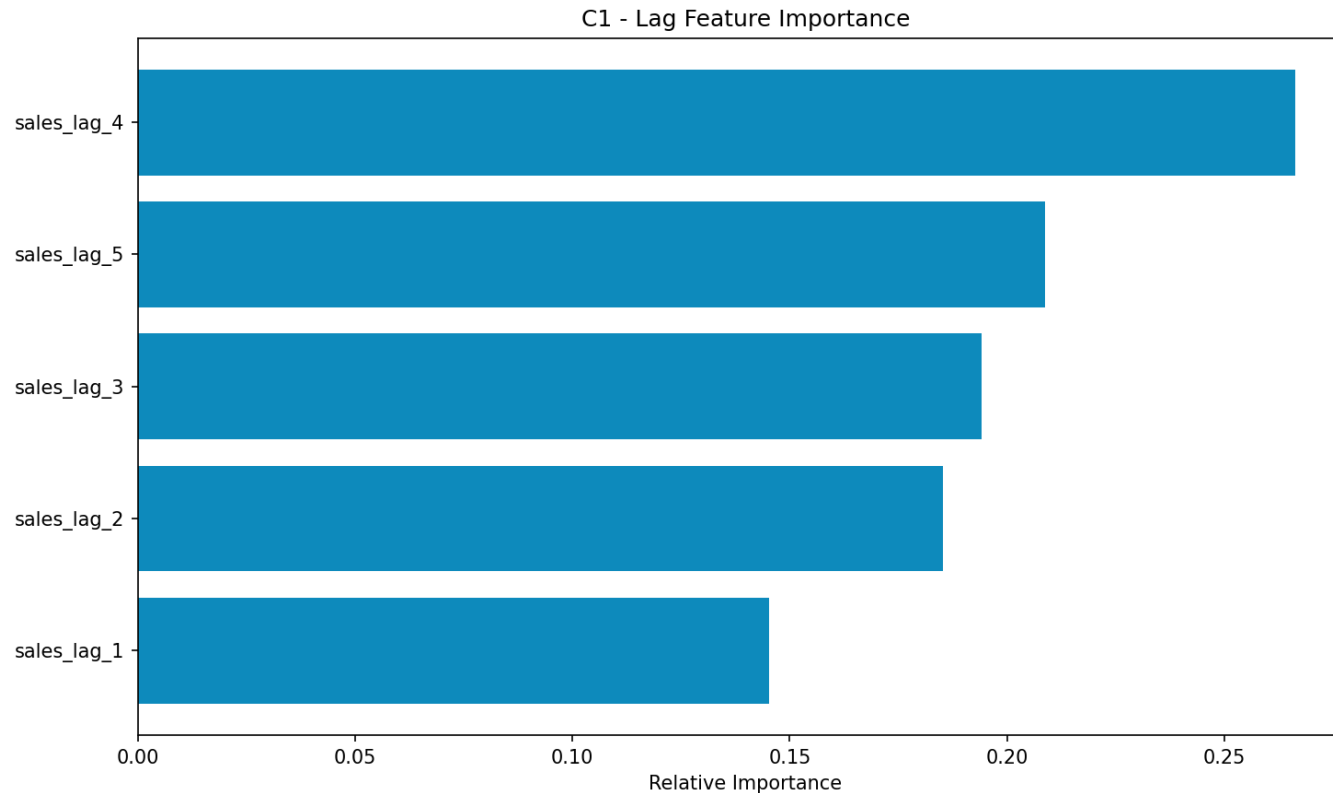
LSTM leads in 3 of 8 categories.

LightGBM leads on average MAE and RMSE.

A deployment model-selection rule still needs an independent validation and test period.

Source: src/evaluation_results/model_metrics.json; minimum MAE within each category

Explainability must name the method actually returned



Saved C1 example

sales_lag_4

0.2664

normalised importance

Method: fallback

lag importance

This output ranks lag importance; it is neither a causal effect nor a local SHAP attribution.

Advanced AI modules are experimental extensions

Module	Available evidence	Interpretation
Neural architecture search	C1 run: 8 architectures evaluated	Small search; scaled metrics
Federated learning	2 simulated clients; 1 FedAvg round	Local demonstration
Meta-learning	Adaptation code; recorded MAML untrained	No demonstrated benchmark gain
Causal analysis	Recorded temporal-feature correlations	Association, not causal proof

Implemented extension does not imply independently validated improvement in forecasting accuracy.

Source: nas_results_C1_1783854913.json; fed_results_C1_iid_1783854914.json; causal_discovery_C1.json; thesis §§4.5,4.8

Application demonstration • embedded offline walkthrough

01 Configure a request

The screenshot shows the PharmaPredictAI web application interface. At the top, there is a navigation bar with the PharmaPredictAI logo, a home icon, and links to Dashboard, Forecasting (highlighted), AI Models, and Analytics. On the right, there is a notification bell with a red '3' and a user profile icon labeled 'AD Admin'. Below the navigation bar, the main content area is titled 'Drug Sales Forecasting' with a subtitle 'Advanced multi-model prediction with uncertainty quantification'. A purple 'Export Results' button is in the top right. The 'Configure Prediction' section contains four fields: 'Drug Category' (a dropdown menu with 'Select Category'), 'Forecast Date' (a date input field with '07/19/2026' and a calendar icon), 'Model Type' (a dropdown menu with 'Ensemble (Recommended)'), and 'Options' (two checkboxes, 'Uncertainty Estimation' and 'Confidence Intervals', both checked). A purple 'Generate Forecast' button is at the bottom left of the configuration area.

Choose a category, date and model in the project interface.

OFFLINE WALKTHROUGH • saved project screens • no narration

Click to play • 45 seconds • saved screens, historical lookup and explanation

Source: Project UI screenshot; C1 historical output chart; saved C1 fallback explanation; 45-second silent MP4

Stacking passes functional and chronological checks

17 checks passed

16 Python tests
1 JavaScript UI test
Chronology + horizon checks
Artifact integrity + errors

Stacking API check

C1 request: 11 Dec 2023
Weekly end: 17 Dec 2023
Forecast: 53.0036
HTTP status: 200

Application output

Actual recorded sales chart
Measured holdout errors
No invented accuracy score
Unsupported intervals omitted

All 8 stacking artifacts produce finite forecasts. The separate holdout measures forecasting error.

Source: tests/test_stacking.py; tests/test_services_smoke.py; tests/test_stacking_ui.cjs; stacking_integration_checks.json

Limitations define the next research steps

Current limitation	Next step
Hybrid data; no source / regional labels	Collect traceable, region-labelled pharmacy histories
Original benchmark horizon / split issues	New stacking is aligned; repeat for the other models
Older API routes use day counts as steps	Stacking uses weeks; extend the fix to older routes
Low sales destabilise percentage error	Add suitable scaled errors and inspect low-volume cases

Operational stock-out reduction, cost savings and real multi-pharmacy generalisation remain unmeasured.

Source: CSV audit; src/evaluation/model_evaluation.py; forecast_service/app/forecasting.py; thesis §§5.6,6.7

Conclusion: a working platform with clear research findings

Delivered

A category-wise forecasting and explanation application.

Observed

Stacking works; simple averaging is better in the new holdout.

Next

Strengthen future-period validation and real pharmacy evaluation.

Selected references

- [1] S. B. Ekanayake, M. Nasmeeen, G. Lakmal, P. Vimanshani and A. Perera. “Predicting medical drug sales in a specific area for categorical drugs using time series forecasting.” SCSE, 2025. Local project copy.
- [2] [G. Ke et al. “LightGBM: A Highly Efficient Gradient Boosting Decision Tree.” Advances in Neural Information Processing Systems 30, 2017, pp. 3146–3154.](#)
- [3] [H. Hewamalage, C. Bergmeir and K. Bandara. “Recurrent Neural Networks for Time Series Forecasting: Current Status and Future Directions.” International Journal of Forecasting, 37\(1\), 2021, pp. 388–427.](#)
- [4] B. R. G. Lakmal. Final-year thesis, CS/2020/015, University of Kelaniya, 2026; accompanying source code, category CSVs and saved evaluation JSON files.

Thank you

Questions & discussion

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